

	Sanjivani Rural Educational Society's SANJIVANI COLLEGE OF ENGINEERING (An Autonomous Institution) Kopergaon – 423 603, Maharashtra.	ACAD-F-15 K
Academic Year: 2025-26	Mini Project – CASE STUDY	Revision: 00 Dated:
Department:	Computer Engineering	Date of Preparation:
Course Code & Name:	Fundamentals of Internet of Things (MDCO223)	Year/Sem: S.Y. Semester-II (Div. -A)

Group ID: 14

Activity Title: Smart Eco-Safe Home Automation System with Emotional OLED Interface

Submitted by:

Roll No	PRN No	Name of the Student	Signature
66	0124UCSM1066	Deokar Sai Sandip	
67	0124UCSM1067	Nere Tejas Vikas	
68	0124UCSM1068	Palande Siddharth Vilas	
69	0124UCSF1069	Gidhad Arti Annasaheb	
70	0124UCSM1070	Rashinkar Sanket Laxman	

Course Incharge

Prof. P. M. Dhanrao

1. Abstract

The rapid advancement of technology has transformed traditional homes into intelligent living spaces through the use of the Internet of Things (IoT). Smart home automation systems are designed to improve convenience, safety, and energy efficiency by enabling devices to monitor surroundings and respond automatically. In today's world, there is an increasing demand for systems that not only automate household functions but also provide real-time alerts and interactive user experiences.

The **Smart Eco-Safe Home Automation System with Emotional OLED Interface** is an innovative IoT-based project developed to create a safer, smarter, and energy-efficient home environment. The system integrates multiple sensors such as IR sensors, gas sensor, rain sensor, and DHT11 sensor with microcontrollers like Arduino Uno and ESP32. These sensors continuously monitor movement, air quality, temperature, humidity, rainfall, and electrical load conditions.

Based on sensor inputs, the system performs automatic actions such as staircase lighting, window control using a servo motor, abnormal load alerts, and air quality management. It also sends real-time notifications through Telegram for remote monitoring.

A unique feature of this project is the **Emotional OLED Interface**, which displays human-like expressions such as alert, rain, bad air, and normal animated eyes. This makes the interaction between humans and machines more engaging and intuitive.

The project is cost-effective, scalable, and suitable for future smart homes. It demonstrates how IoT can be combined with automation and emotional interfaces to create intelligent systems for modern living.

2. Introduction

In modern society, homes are becoming smarter with the help of automation and connected devices. Traditional home systems require manual control of appliances, lighting, ventilation, and safety monitoring. This often leads to energy wastage, delayed response to emergencies, and inconvenience for users. To overcome these limitations, IoT-based smart home systems have emerged as a practical solution.

The **Smart Eco-Safe Home Automation System with Emotional OLED Interface** is designed to automate household functions while ensuring safety, environmental awareness, and real-time communication. The project uses sensors to monitor important household conditions such as motion detection, gas leakage, air quality, rainfall, temperature, humidity, and abnormal electrical load.

The system uses **Arduino Uno** as the control unit for reading sensor data and decision-making. An **ESP32** is used as the smart interface layer for OLED display control and Telegram notifications. Together, they create a dual-controller architecture that improves efficiency and modularity.

This system includes automatic staircase lighting using IR sensors, rain-based window closing using a servo motor, gas and air quality monitoring, fan load anomaly detection, and emotional UI feedback through OLED expressions.

The project is highly useful in homes, offices, hostels, and smart buildings. It enhances comfort, reduces energy consumption, increases safety, and introduces a friendly emotional interaction between the user and the system.

3. System Design

The system is divided into two major layers:

1. Control Layer (Arduino Uno)

This layer is responsible for:

- Reading all sensor values
- Detecting environmental changes
- Making logical decisions
- Sending HIGH/LOW signals to ESP32

2. Smart Interface Layer (ESP32)

This layer is responsible for:

- Receiving signals from Arduino
- Updating OLED display
- Sending Telegram notifications
- Managing emotional user interface

Input Devices

- IR Sensors (2) – motion and direction detection
- MQ Gas Sensor – gas/air quality monitoring
- Rain Sensor – rainfall detection
- DHT11 – temperature and humidity
- Fan Load Sensor – abnormal load detection

Output Devices

- OLED Display
- Servo Motor
- Buzzer
- Staircase Lights
- Telegram Alerts

System Flow

Sensors → Arduino → ESP32 → OLED + Telegram + Automation Actions

The modular design makes the system efficient, flexible, and expandable for future upgrades.

4. Explanation of each Module

4.1 Arduino UNO



Description

- Arduino Uno is a popular microcontroller board based on the ATmega328P processor.
- It is widely used in automation, robotics, and IoT projects.
- The board contains digital and analog input/output pins for connecting sensors and devices.
- It operates at 5V and can be programmed using Arduino IDE.
- It acts as the main control unit of the project.

Purpose

- To collect input signals from all connected sensors.
- To process sensor data using programmed logic.
- To make decisions for automation tasks.
- To control outputs such as buzzer, lights, and signals to ESP32.

Function

- Reads data from IR sensors, gas sensor, rain sensor, and DHT11.
- Checks conditions like motion, gas level, rain detection, and load anomalies.
- Executes automation logic based on sensor readings.
- Sends HIGH/LOW digital signals to ESP32.
- Controls local devices such as lights and buzzer.

4.2 ESP32 Development Board



Description

- ESP32 is an advanced microcontroller with built-in Wi-Fi and Bluetooth.
- It is faster and more powerful than basic controllers.
- It supports internet connectivity for IoT applications.
- It works on 3.3V logic levels.
- ESP32 can communicate with displays, sensors, and cloud platforms.

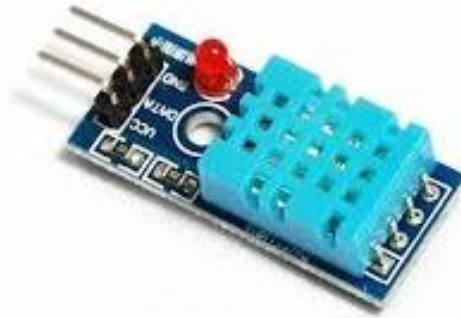
Purpose

- To provide wireless connectivity.
- To manage OLED display output.
- To send alerts through Telegram.
- To receive signals from Arduino Uno.

Function

- Receives control signals from Arduino.
- Updates OLED display with messages and expressions.
- Connects to Wi-Fi network.
- Sends notifications to users through Telegram.

4.3 DHT11 Sensor (Temperature and Humidity Sensor)



Description

- The DHT11 is a low-cost digital sensor used for measuring temperature and humidity.
- It contains a humidity sensing component and a thermistor.
- It provides calibrated digital output.
- It operates on 3V to 5V supply.
- It has three pins: VCC, GND, and DATA.
- It is widely used in smart home and environmental monitoring systems.

Purpose

- To measure room temperature accurately.
- To monitor humidity levels in the environment.
- To provide real-time environmental data.
- To support air quality decisions.

Function

- Detects humidity using a moisture-sensitive capacitor.
- Measures temperature using a thermistor.
- Converts analog readings into digital data.
- Sends data to Arduino.
- System uses values for monitoring and automation.

4.4 SSD1306 OLED Display



Description

- SSD1306 OLED is a compact graphical display module.
- It communicates using I2C protocol.
- It offers clear text and graphics output.
- It consumes low power.
- It is widely used in embedded projects.

Purpose

- To display system status messages.
- To show emotional expressions.
- To provide real-time alerts visually.
- To make the system interactive.
- To improve user experience.

Function

- Receives commands from ESP32.
- Displays normal messages like System Active.
- Shows animated eyes in idle mode.
- Displays alert screens during events.
- Updates content continuously in real time.

5. Working

A. System Initialization

- The system starts when power is supplied to the circuit.
- Arduino Uno and ESP32 initialize all connected components.
- Sensors such as IR sensors, DHT11, MQ gas sensor, and rain sensor are activated.
- OLED display starts and shows system startup message.
- Communication between Arduino and ESP32 is established.
- Wi-Fi connection is initialized for Telegram notifications.

B. Data Collection from Sensors

- All sensors continuously monitor surrounding conditions.
- The IR sensors detect human movement and direction on stairs.
- The DHT11 sensor measures: Temperature and Humidity.
- The rain sensor detects:
 - o Presence of water droplets
 - o Rain / No Rain condition
- The MQ gas sensor detects:
 - o Harmful gases in air
 - o Air quality level
- Load monitoring input detects:
 - o Normal load
 - o Abnormal fan load condition.

C. Signal Transmission to Controller

- Sensors send their outputs to Arduino Uno.
- Signals may be:
 - o Digital (IR sensors, rain sensor, DHT11)
 - o Analog (MQ gas sensor, load sensor)

-
- Analog signals are converted into digital values using ADC.
 - Each sensor is connected to specific input pins.
 - Arduino processes signals and sends HIGH/LOW outputs to ESP32.

D. Data Processing

- Arduino reads all sensor values continuously.
- Raw data is processed using programmed logic.
- Values are converted into meaningful information:
 - o Temperature → Degree Celsius
 - o Humidity → Percentage
 - o Gas Level → Air Quality Status
 - o Rain Status → Rain / No Rain
 - o Motion Input → Stair Direction

E. Automatic Staircase Lighting

- Two IR sensors are used for motion-based staircase lighting.
- If a person moves upward, system detects UP direction.
- If a person moves downward, system detects DOWN direction.
- Stair lights turn ON sequentially according to direction.
- After a delay, lights turn OFF automatically.
- This reduces unnecessary energy consumption.

F. Rain Detection and Window Control

- Rain sensor continuously checks for rainfall.
- If rain is detected:
 - o Servo motor closes the window
 - o Telegram alert is sent
 - o OLED shows rain alert mode
- If rain stops, system returns to normal monitoring state.

G. Air Quality Monitoring and Ventilation

- MQ gas sensor monitors indoor air quality.
- DHT11 provides supporting environmental data.
- If air quality becomes poor:
 - o Window opens automatically using servo motor
 - o OLED shows bad air warning
 - o Telegram notification is sent
- If air quality becomes normal:
 - o Window closes automatically

H. Fan Load Detection and Safety Alert

- System checks electrical load condition of fan or connected device.
- If abnormal load is detected:
 - o Buzzer turns ON
 - o Telegram warning message is sent
 - o OLED shows alert mode
- If load returns normal, alert stops automatically.

I. Emotional OLED Display Operation

- OLED works as the smart visual interface.
- During normal condition, display alternates between:
 - o Animated robotic eyes
 - o System status messages

J. IoT Communication (Telegram Alerts)

- ESP32 connects to Wi-Fi network.
- Important events are sent to users through Telegram.
- Alerts include:

-
- o Rain detected
 - o Poor air quality
 - o Abnormal load condition

K. Continuous Monitoring Loop

- The system works continuously in a loop.
- Repeated operations include:
 - o Read sensor data
 - o Process conditions
 - o Trigger actions
 - o Update OLED display
 - o Send alerts if required

L. Power Supply and Stability

- The system uses regulated power supply for all components.
- Arduino works at 5V and ESP32 works at 3.3V.
- Voltage divider is used for safe signal conversion.
- Stable power ensures reliable performance.
- USB adapter or external supply can be used.

M. Error Handling and Maintenance

- Sensors may produce noise or false readings.
- Basic filtering logic improves accuracy.
- Components should be checked regularly.
- Dust or moisture may affect sensors over time.
- Proper maintenance ensures long life of system.

6. Implementation

Code:

// arduino code :

```
#include <Servo.h>
```

```
#include <DHT.h>
```

```
#define DHTPIN 2
```

```
#define DHTTYPE DHT11
```

```
DHT dht(DHTPIN, DHTTYPE);
```

```
// ===== SERVOS =====
```

```
Servo windowServo; // Air system
```

```
Servo rainServo; // Rain system
```

```
// ===== PINS =====
```

```
int gasPin = A0;
```

```
int airSignalPin = 7; // air signal
```

```
int rainPin = 3;
```

```
int rainSignalPin = 13; // rain signal
```

```
// ===== STATES =====
```

```
int windowState = 0;
```

```
bool rainState = false;
```

```
// ===== THRESHOLDS =====
```

```
int gasOpenThreshold = 650;
```

```
int gasCloseThreshold = 500;
```

```
// ===== TIMING =====
```

```
unsigned long lastBadAirTime = 0;
```

```
unsigned long closeDelay = 4000;
```

```
void setup() {
```

```
    Serial.begin(9600);
```

```
    // Pins
```

```
    pinMode(gasPin, INPUT);
```

```
    pinMode(airSignalPin, OUTPUT);
```

```
    pinMode(rainPin, INPUT);
```

```
    pinMode(rainSignalPin, OUTPUT);
```

```
    dht.begin();
```

```
    // Servos
```

```
    windowServo.attach(9);
```

```
    rainServo.attach(4);
```

```
    windowServo.write(0);
```

```
    rainServo.write(0);
```

```
}
```

```
void loop() {
```

```
    // ===== ☁ RAIN SYSTEM =====
```

```
    int rain = digitalRead(rainPin);
```

```
    if (rain == LOW && !rainState) {
```

```
        Serial.println("☁ Rain Detected");
```

```
        rainServo.write(90);
```

```
        digitalWrite(rainSignalPin, HIGH);
```

```
        rainState = true;
```

```
}
```

```
else if (rain == HIGH && rainState) {
```

```
    Serial.println("☀ Rain Stopped");
```

```
    rainServo.write(0);
```

```
    digitalWrite(rainSignalPin, LOW);
```

```
    rainState = false;
```

```
}
```

```
// ===== ✨ AIR SYSTEM =====
```

```
int gas = 0;
```

```
for (int i = 0; i < 7; i++) {
```

```
    gas += analogRead(gasPin);
```

```
    delay(30); // (same as tera original)
```

```
}
```

```
gas = gas / 7;
```

```
float temp = dht.readTemperature();
```

```
float hum = dht.readHumidity();
```

```
if (isnan(temp) || isnan(hum)) {
```

```
    Serial.println("ERROR");
```

```
    delay(1500);
```

```
    return;
```

```
}
```

```
Serial.print("GAS:");
```

```
Serial.print(gas);
```

```
Serial.print(",TEMP:");
```

```
Serial.print(temp);
```

```
Serial.print(",HUM:");
Serial.println(hum);

bool badAir = false;

if (gas > gasOpenThreshold || (temp > 32 && hum > 70)) {
    badAir = true;
    lastBadAirTime = millis();
}

// 🪟 OPEN
if (badAir && windowState == 0) {
    windowServo.write(90);
    digitalWrite(airSignalPin, HIGH);
    windowState = 1;

    Serial.println("STATUS:OPEN");
}

// ✅ CLOSE
if (!badAir && windowState == 1) {
    if (gas < gasCloseThreshold && millis() - lastBadAirTime > closeDelay) {
        windowServo.write(0);
        digitalWrite(airSignalPin, LOW);
        windowState = 0;

        Serial.println("STATUS:CLOSE");
    }
}

delay(1000); // same as tera code
}
```


7. Output

Telegram Chat-bot for Smart & Safe Home Automation :



The **Smart Eco-Safe Home Automation System with Emotional OLED Interface** was successfully implemented and tested under different conditions. The system accurately responded to sensor inputs and performed automation tasks in real time. It effectively controlled household functions, improved safety, and provided interactive feedback through the OLED display and Telegram alerts.

The **automatic staircase lighting system** worked efficiently using IR sensors. When movement was detected, the lights turned ON sequentially according to direction and turned OFF automatically after a delay, helping to save electricity.

The **rain detection system** successfully detected rainfall and automatically closed the window using a servo motor. At the same time, Telegram alerts were sent and the OLED display showed a rain warning message.

The **air quality monitoring system** detected harmful gases using the MQ sensor. When poor air quality was sensed, the window opened automatically for ventilation and warning messages were displayed. When conditions became normal, the window closed again.

The **fan load detection system** identified abnormal load conditions and activated the buzzer, OLED alert mode, and Telegram notifications. This improved safety by warning users about possible electrical issues.

Overall, the project successfully achieved its objectives of automation, safety, energy efficiency, and smart user interaction. It is a practical and cost-effective solution for future smart homes.

8. Conclusion

The **Smart Eco-Safe Home Automation System with Emotional OLED Interface** is an innovative IoT-based project developed to improve modern living by combining automation, safety, environmental monitoring, and user interaction in one smart system. It demonstrates how technology can reduce manual effort, increase convenience, and create a safer and more energy-efficient home environment.

One of the key achievements of this project is intelligent automation of household tasks. Features such as automatic staircase lighting based on motion detection help save electricity by turning lights ON only when required. The automated window control system responds to rain and poor air quality conditions, making the home smarter and eco-friendly.

The project also focuses strongly on safety. It continuously monitors abnormal fan load conditions, gas leakage risks, and changes in air quality. Whenever any dangerous condition is detected, the buzzer is activated, Telegram alerts are sent instantly, and warning messages are shown on the OLED display. This real-time alert system helps users take quick action and avoid possible accidents.

A unique highlight of the project is the **Emotional OLED Interface**. Instead of displaying only text, the OLED screen shows animated robotic eyes, status symbols, and different alert expressions for rain, danger, and poor air quality. This creates a more engaging and human-like interaction between the user and the system.

The use of both Arduino Uno and ESP32 provides an efficient architecture where Arduino handles sensor data and control logic, while ESP32 manages display functions and IoT communication. This makes the system modular, reliable, and easy to upgrade in the future.

In conclusion, this project is more than a basic automation system. It is a smart, safe, and interactive home assistant that shows the real potential of IoT in everyday life. It improves comfort, promotes energy saving, enhances safety, and represents a practical step toward future smart homes.

9. References

1. Adeel Javed, *Building Arduino Projects for the Internet of Things*, Apress, 2016.
Link: <https://link.springer.com/book/10.1007/978-1-4842-1940-9>
2. Charles Bell, *Beginning Sensor Networks with Arduino and Raspberry Pi*, Apress, 2014.
Link: <https://link.springer.com/book/10.1007/978-1-4302-5825-4>
3. Arduino Official Website
Link: <https://www.arduino.cc>
4. ESP32 Documentation (Espressif)
Link: <https://docs.espressif.com>
5. Wokwi Online Simulator (Arduino & ESP32 Simulation)
Link: <https://wokwi.com>
6. Tinkercad Circuits (Circuit Design & Simulation)
Link: <https://www.tinkercad.com>
7. Random Nerd Tutorials (ESP32 & IoT Guides)
Link: <https://randomnerdtutorials.com>
8. Telegram Bot API Documentation
Link: <https://core.telegram.org/bots/api>

10. Group Photo:

